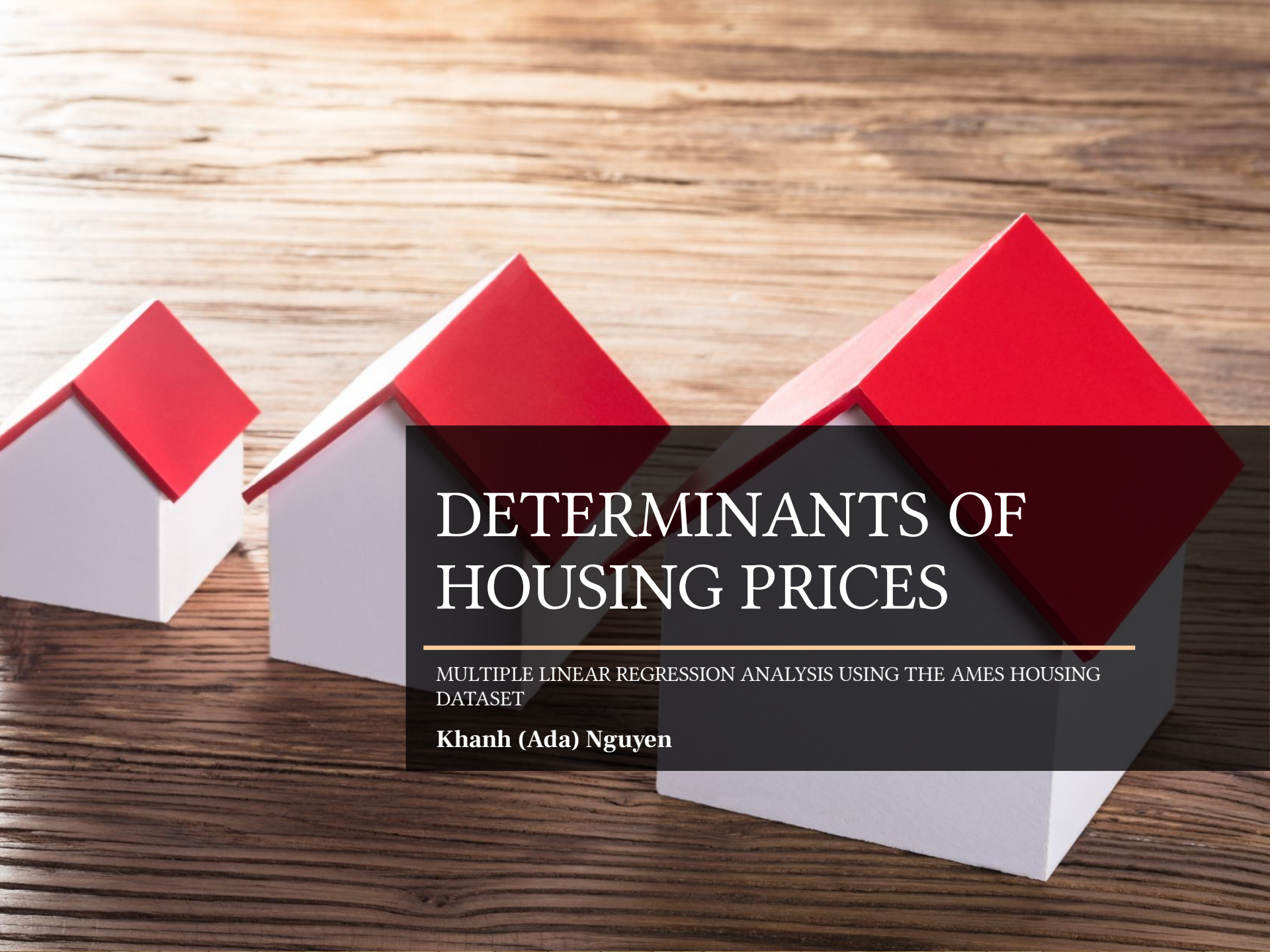


The background of the slide features three white, three-dimensional house models with red roofs, arranged in a row on a dark wooden surface. The houses increase in size from left to right. A dark, semi-transparent rectangular box is overlaid on the right side of the image, containing the title and subtitle text.

DETERMINANTS OF HOUSING PRICES

MULTIPLE LINEAR REGRESSION ANALYSIS USING THE AMES HOUSING
DATASET

Khanh (Ada) Nguyen

AGENDA

01	Introduction
02	Data Preparation
03	Exploratory Data Analysis
04	Model building
05	Model Diagnostic
06	Influence Analysis
07	Prediction and Parameter interpretation

INTRODUCTION

- **Housing prices** depend on structural, locational, and quality-related factors
- **Goal:** identify key determinants of housing prices
- **Data: Ames Housing, Iowa (2006-2010)**
 - Train set: 1,460 observations $\times$ 80 variables (including response variable)
 - Test set: 1,459 observations $\times$ 79 variables
 - Response variable: *SalePrice*
- **Dataset Link:**
 - Kaggle <https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/overview>



Which characteristics are most strongly associated with sale price?



How do structural features affect housing value?



What role does neighborhood/location play?



Can multiple linear regression explain price variation?



How can we predict housing prices and quantify uncertainty?

RESEARCH QUESTIONS

DATA PREPARATION OVERVIEW

1

Handling
missing values
based on their
meaning

2

Processing
categorical
variables

3

Conducting
exploratory
data analysis
(EDA)

HANDLING MISSING VALUES

19 variables contained missing values

Categorical Missing Values <i>absence of feature rather than missing data</i>	16	• No pool, no alley, no fence (<i>PoolQC, Alley, Fence</i>) • No garage (<i>GarageType, GarageFinish, GarageQual, GarageCond</i>) • No basement (<i>BsmtExposure, BsmtFinType1, BsmtFinType2, BsmtQual, BsmtCond</i>) • No fireplace (<i>FireplaceQu</i>) • No masonry veneer (<i>MasVnrType</i>) → All missing values were recoded as "None" • MiscFeature → dropped due to high missingness and limited usefulness • Single missing value on <i>Electrical</i> → imputed with mode
Numeric Missing Values	3	• LotFrontage → imputed by Neighborhood median • MasVnrArea and GarageYrBlt → replaced with 0

PROCESSING CATEGORICAL VARIABLES

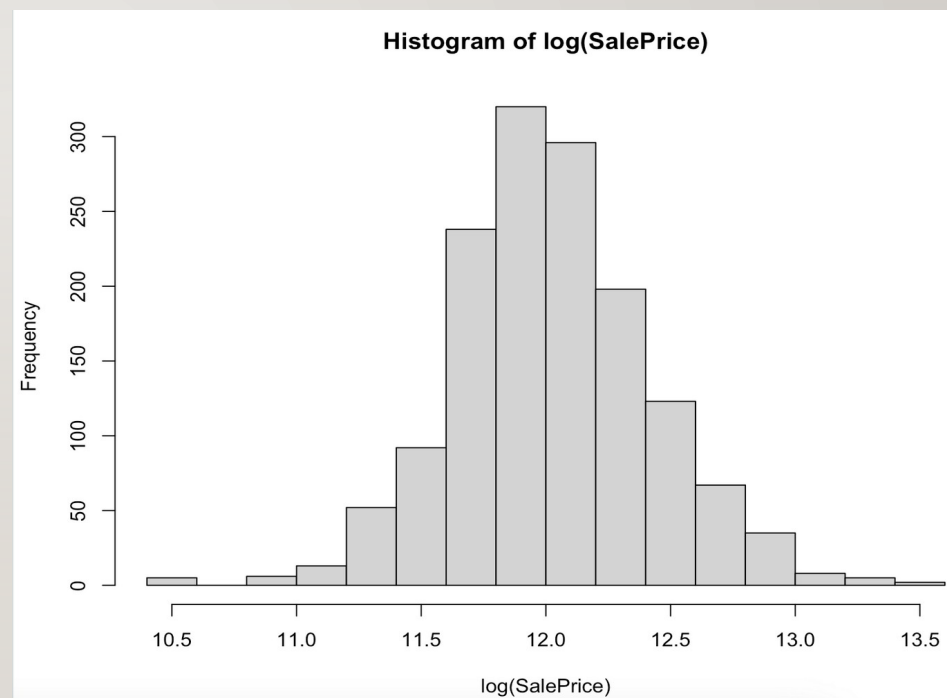
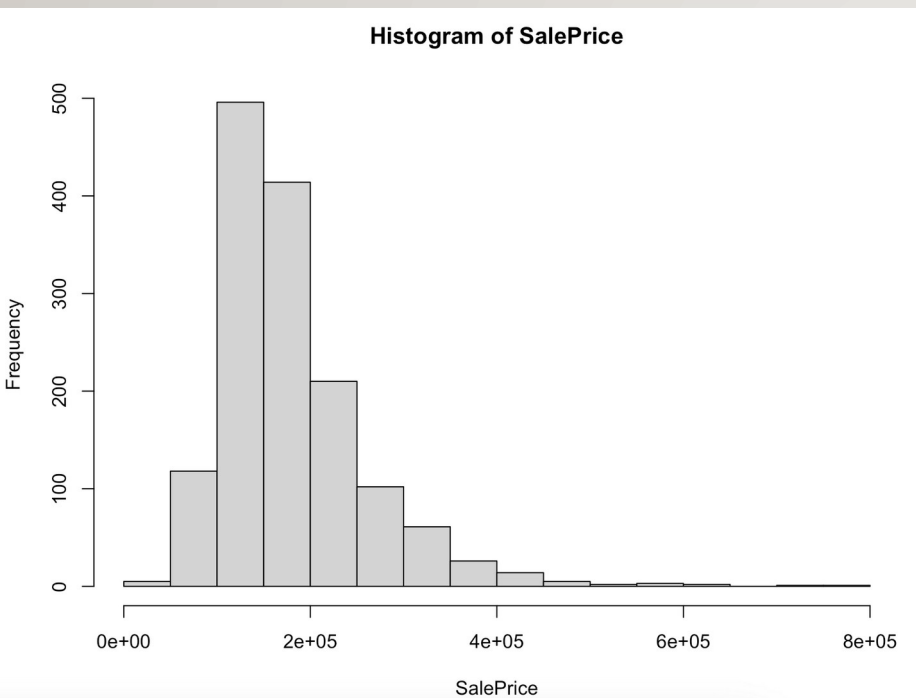
- 42 categorical variables identified
- Focus on 19 variables with > 5 levels

Keep as-is (9 variables)	Meaningful categories	Neighborhood, HouseStyle, SaleType, SaleCondition, RoofStyle, RoofMatl, Foundation, Heating, Functional
Preserve "None" as a separate category (11 variables)	Represents absence of feature	FireplaceQu, GarageType, GarageFinish, GarageQual, GarageCond, BsmtExposure, BsmtFinType1, BsmtFinType2, BsmtQual, BsmtCond, MasVnrType
Rare categories grouped into 'Other' (4 variables)	Categories with less than 2% frequency	Exterior1st, Exterior2nd, Condition1, Condition2

CONDUCTED EXPLORATORY DATA ANALYSIS (EDA)

Response Variable Transformation

- SalePrice was highly right-skewed → Log transformation



CORRELATION ANALYSIS

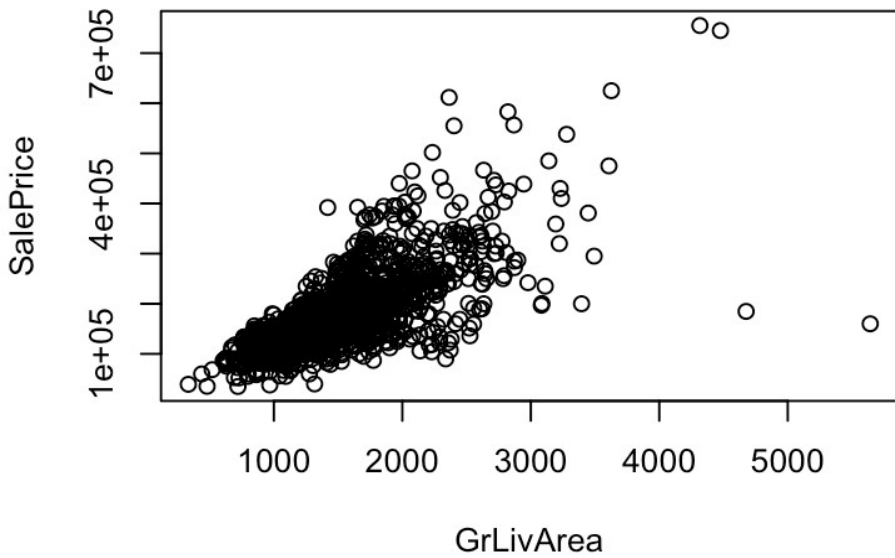
- Strong predictors: OverallQual, GrLivArea, GarageCars, GarageArea, TotalBsmtSF

OverallQual	GrLivArea	GarageCars	GarageArea	TotalBsmtSF	X1stFlrSF	FullBath	TotRmsAbvGrd
0.7909816	0.7086245	0.6404092	0.6234314	0.6135806	0.6058522	0.5606638	0.5337232
YearBuilt	YearRemodAdd						
0.5228973	0.5071010						

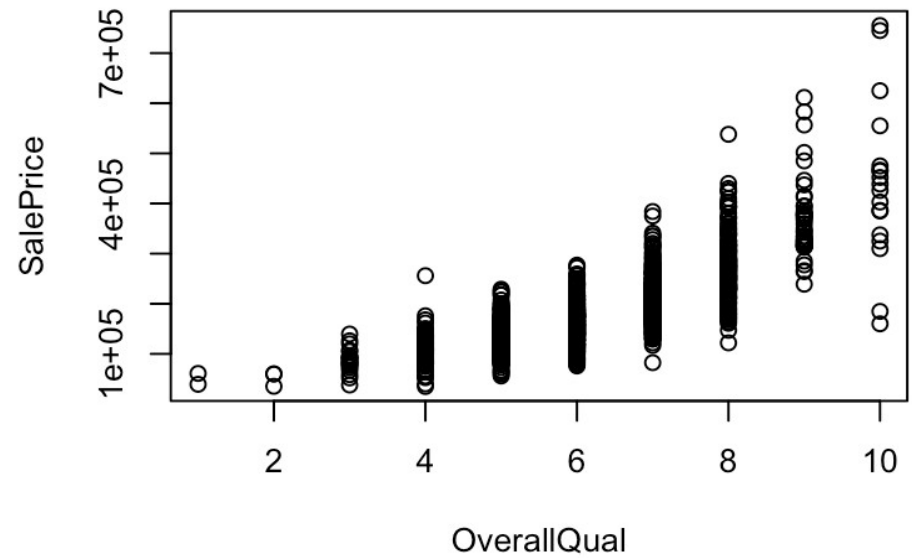
- Moderate to high correlations among some predictors (GarageCars and GarageArea , GrLivArea and TotRmsAbvGrd) → potential multicollinearity

	SalePrice	OverallQual	GrLivArea	GarageCars	GarageArea	TotalBsmtSF	X1stFlrSF	FullBath
SalePrice	1.00	0.79	0.71	0.64	0.62	0.61	0.61	0.56
OverallQual	0.79	1.00	0.59	0.60	0.56	0.54	0.48	0.55
GrLivArea	0.71	0.59	1.00	0.47	0.47	0.45	0.57	0.63
GarageCars	0.64	0.60	0.47	1.00	0.88	0.43	0.44	0.47
GarageArea	0.62	0.56	0.47	0.88	1.00	0.49	0.49	0.41
TotalBsmtSF	0.61	0.54	0.45	0.43	0.49	1.00	0.82	0.32
X1stFlrSF	0.61	0.48	0.57	0.44	0.49	0.82	1.00	0.38
FullBath	0.56	0.55	0.63	0.47	0.41	0.32	0.38	1.00
TotRmsAbvGrd	0.53	0.43	0.83	0.36	0.34	0.29	0.41	0.55
YearBuilt	0.52	0.57	0.20	0.54	0.48	0.39	0.28	0.47
YearRemodAdd	0.51	0.55	0.29	0.42	0.37	0.29	0.24	0.44

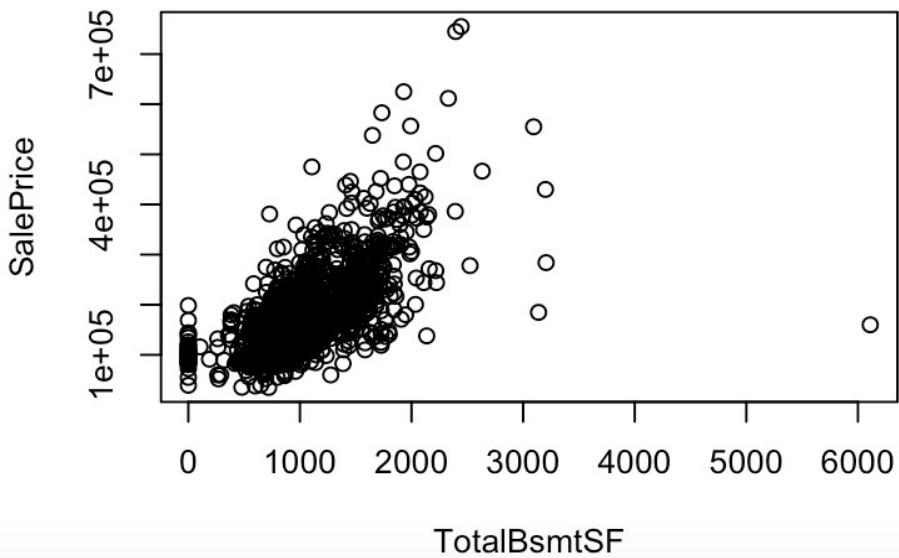
SalePrice vs GrLivArea



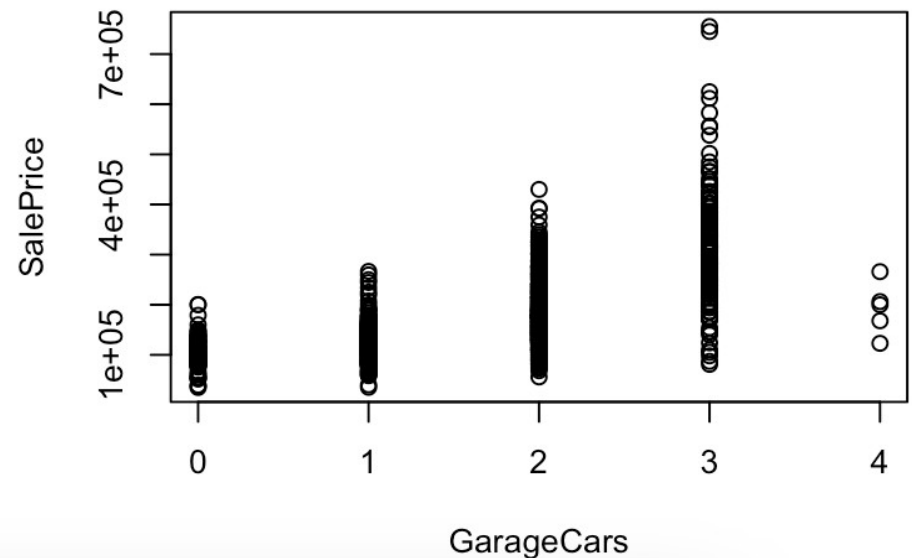
SalePrice vs OverallQual



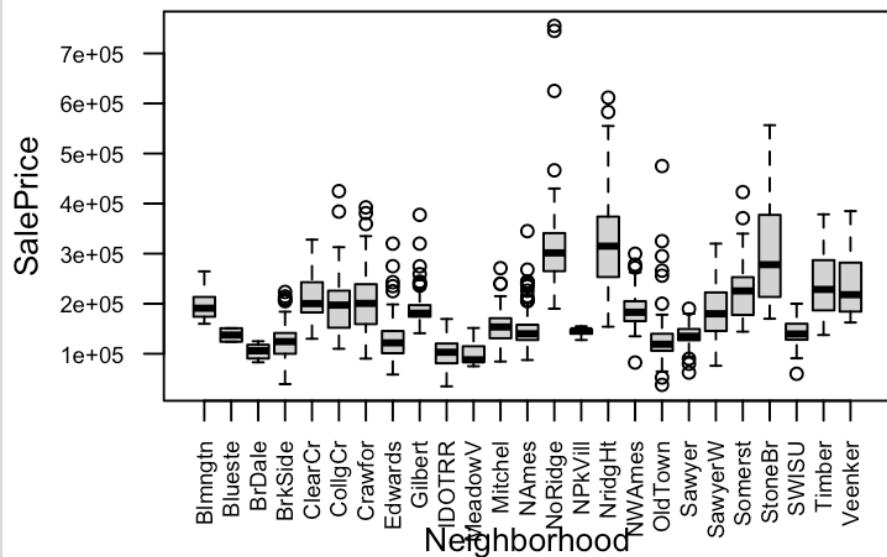
SalePrice vs TotalBsmtSF



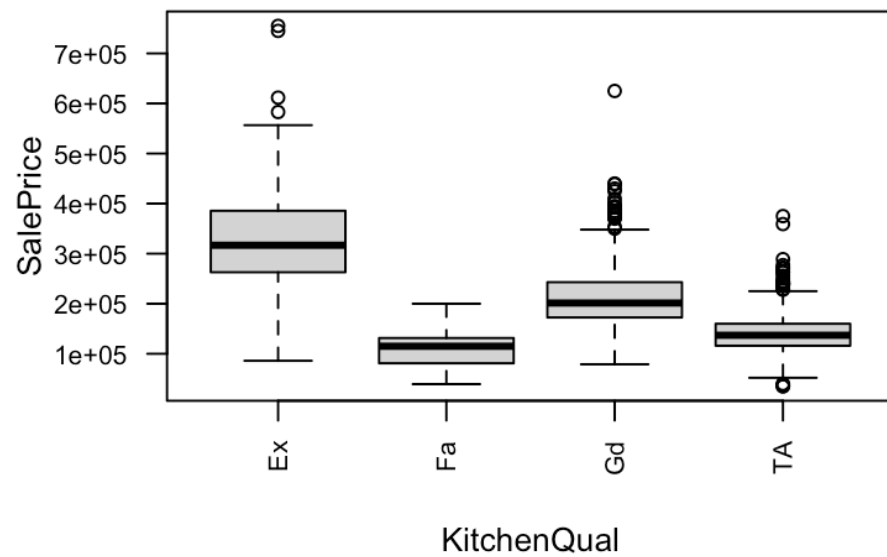
SalePrice vs GarageCars



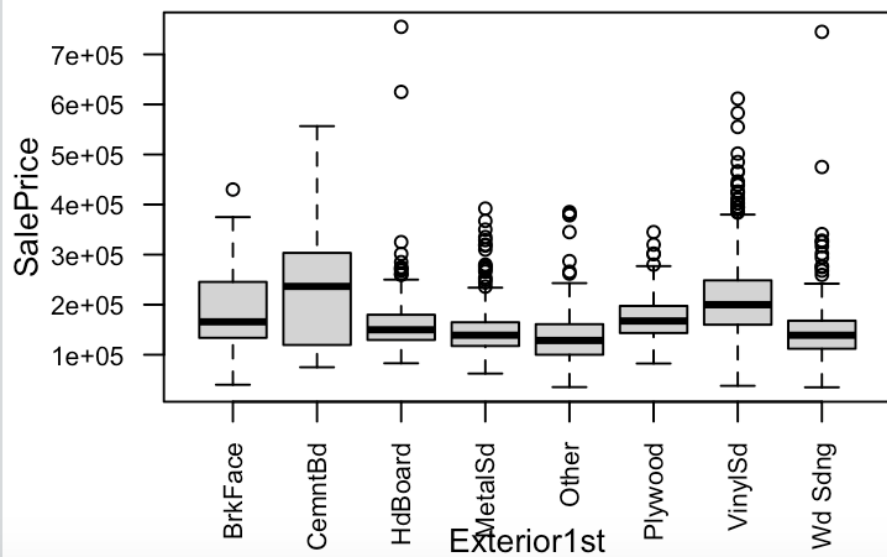
SalePrice by Neighborhood



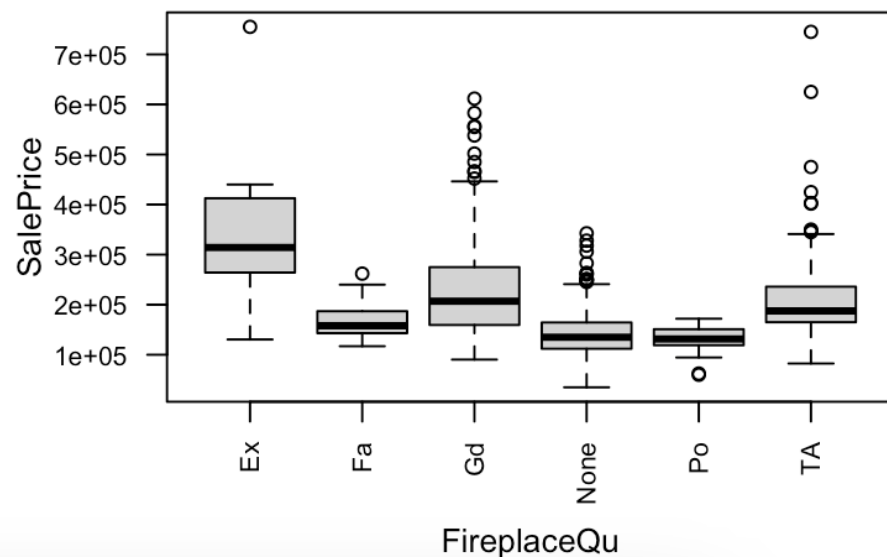
SalePrice by KitchenQual



SalePrice by Exterior1st



SalePrice by Fireplace Quality



FINAL OUTPUT

- Cleaned dataset exported as
ames_housing_train_cleaned.csv
- No missing values and proper variable formatting
- Dataset ready for regression modeling
- Next step: Model fitting and Diagnostics

MODEL BUILDING

Variables of Interest

- Overall Quality
- Above Ground Living Area
- Garage Cars / Garage Area
- Total Basement Square Feet
- Full Bath Count
- Year Built
- Kitchen Quality
- Lot Area

High Correlation with Sale Price

variable	correlation
OverallQual	0.8171844
GrLivArea	0.7009267
GarageCars	0.6806248
GarageArea	0.6508876
TotalBsmtSF	0.6121340
X1stFlrSF	0.5969810
FullBath	0.5947705
YearBuilt	0.5865702
YearRemodAdd	0.5656078

MODEL BUILDING

Model 1: Base Model With Highest Correlations

```
> model1 <- lm(SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF, data = dat_clean)
> summary(model1) # decent baseline with 0.7942 R-squared
```

Call:

```
lm(formula = SalePrice ~ OverallQual + GrLivArea + GarageCars +
    TotalBsmtSF, data = dat_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-469856	-19956	-1360	17200	286304

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-99248.853	4639.866	-21.39	<2e-16	***
OverallQual	23572.236	1072.465	21.98	<2e-16	***
GrLivArea	45.643	2.484	18.38	<2e-16	***
GarageCars	18582.209	1747.412	10.63	<2e-16	***
TotalBsmtSF	32.520	2.838	11.46	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 38920 on 1455 degrees of freedom

Multiple R-squared: 0.7607, Adjusted R-squared: 0.76

F-statistic: 1156 on 4 and 1455 DF, p-value: < 2.2e-16

MODEL BUILDING

Model 2: Adding a Group of Predictors

Model 2

$$\begin{aligned} \text{PredictedSalePrice} = & - 477400 + 13360(\text{OverallQual}) + 34.34(\text{GrLivArea}) + \\ & 11230(\text{GarageCars}) + 17.04(\text{TotalBsmtSF}) + 252.5(\text{YearBuilt}) + 0.620(\text{LotArea}) - \\ & 5.792(\text{FullBath}) - 6667(\text{NeighborhoodBlueste}) - 12340(\text{NeighborhoodBrDale}) + \\ & 15760(\text{NeighborhoodBrkSide}) + 23610(\text{NeighborhoodClearCr}) + \\ & 144900(\text{NeighborhoodColgCr}) + 40850(\text{NeighborhoodCrawfor}) + \\ & 386.8(\text{NeighborhoodEdwards}) + 10220(\text{NeighborhoodGilbert}) + \\ & 1095(\text{NeighborhoodIDOTRR}) - 5223(\text{NeighborhoodMeadowV}) + \\ & 8141(\text{NeighborhoodMitchel}) + 10080(\text{NeighborhoodNAMES}) + \\ & 71780(\text{NeighborhoodNoRidge}) - 14000(\text{NeighborhoodNPkVill}) + \\ & 52310(\text{NeighborhoodNridgHt}) + 7798(\text{NeighborhoodNWAmes}) - \\ & 1671(\text{NeighborhoodOldTown}) + 11320(\text{NeighborhoodSawyer}) + \\ & 10810(\text{NeighborhoodSawyerW}) + 20290(\text{NeighborhoodSomerset}) + \\ & 62420(\text{NeighborhoodStoneBr}) + 37210(\text{NeighborhoodSWISU}) + \\ & 22230(\text{NeighborhoodTimber}) + 43410(\text{NeighborhoodVeenker}) - \\ & 54490(\text{KitchenQualFa}) - 45680(\text{KitchenQualGd}) - 53320(\text{KitchenQualTA}) \end{aligned}$$

MODEL BUILDING

Model 3, 4, and Model Final 1

- Interaction Terms
- Polynomials
- Backward Elimination with Aikake Information Criteria
 - FullBath, OverallQual:GarageCars, $I(\text{GarageCars}^2)$, $I(\text{YearBuilt}^2)$, $I(\text{GrLivArea}^2)$ were all removed from the model
- ANOVA Comparison

```
> anova(model1, model2, model3, model4, model_final)
Analysis of Variance Table

Model 1: SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF
Model 2: SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF +
  YearBuilt + Neighborhood + FullBath + KitchenQual + LotArea
Model 3: SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF +
  YearBuilt + Neighborhood + FullBath + KitchenQual + LotArea +
  OverallQual:GrLivArea + OverallQual:GarageCars + OverallQual:TotalBsmtSF +
  GrLivArea:GarageCars + GrLivArea:TotalBsmtSF
Model 4: SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF +
  YearBuilt + Neighborhood + FullBath + KitchenQual + LotArea +
  OverallQual:GrLivArea + OverallQual:GarageCars + OverallQual:TotalBsmtSF +
  GrLivArea:GarageCars + GrLivArea:TotalBsmtSF + I(GrLivArea^2) +
  I(LotArea^2) + I(YearBuilt^2) + I(GarageCars^2)
Model 5: SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF +
  YearBuilt + Neighborhood + KitchenQual + LotArea + I(LotArea^2) +
  OverallQual:GrLivArea + OverallQual:TotalBsmtSF + GrLivArea:GarageCars +
  GrLivArea:TotalBsmtSF
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	1455	2.2037e+12				
2	1425	1.5661e+12	30	6.3765e+11	26.9911	<2e-16 ***
3	1420	1.1210e+12	5	4.4509e+11	113.0414	<2e-16 ***
4	1416	1.1151e+12	4	5.9393e+09	1.8855	0.1105
5	1421	1.1172e+12	-5	-2.1241e+09	0.5395	0.7465

```
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

MODEL BUILDING

Model Evaluation Metrics

Model	Adj_R2	AIC	BIC
model1	0.76001	35012.38	35044.1
model2	0.82586	34573.69	34763.99
model3	0.874912	34095.53	34312.26
model4	0.875223	34095.77	34333.65
model_final	0.875426	34088.55	34300

MODEL DIAGNOSTIC

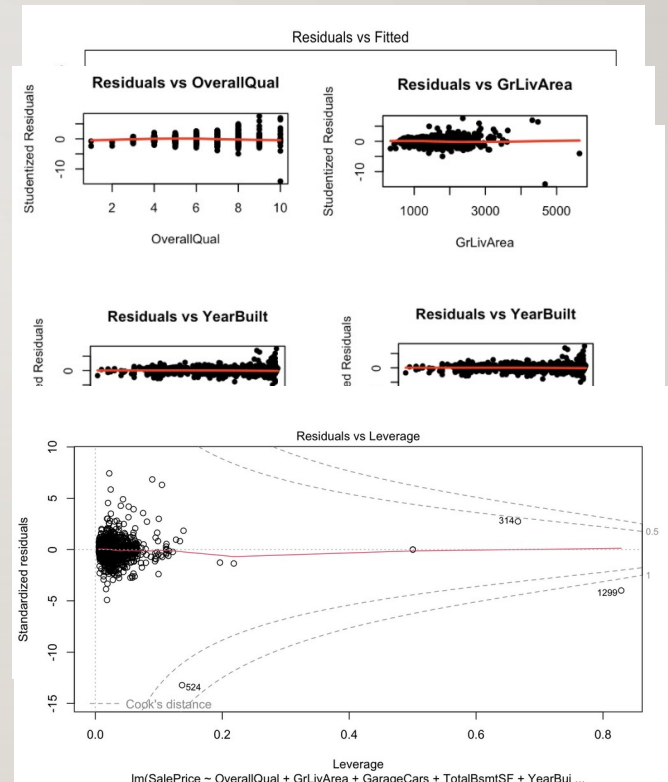
Residuals vs. Fitted plot: non-constant variance and right-skew with extreme outliers (899, 692, 524)

Q-Q plot: heavy tails because of extreme outliers.

Residuals against each predictor:

- Curve up in GrLivArea and TotalBsmtSF
- OverallQual: non-linear trend
- YearBuilt and OverallQual: megaphone shape

Residuals vs. Leverage: 314, 524, and 1299, with high Cook's Distance



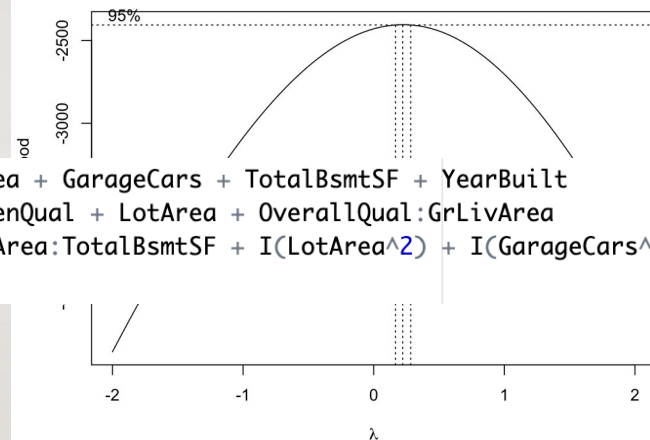
Box-Cox transformation

Lambda is suggested at 0.22.

Hence, we use log

transformation on SalePrice

```
bcox <- boxcox(SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF + YearBuilt  
+ Neighborhood + FullBath + KitchenQual + LotArea + OverallQual:GrLivArea  
+ OverallQual:TotalBsmtSF + GrLivArea:TotalBsmtSF + I(LotArea^2) + I(GarageCars^2),  
lambda = seq(-2, 2, length = 20))
```



```
model_final_2 <- lm(log10(SalePrice) ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF  
+ YearBuilt + Neighborhood + FullBath + KitchenQual + LotArea  
+ OverallQual:GrLivArea + OverallQual:TotalBsmtSF + GrLivArea:TotalBsmtSF  
+ I(LotArea^2) + I(GarageCars^2))
```

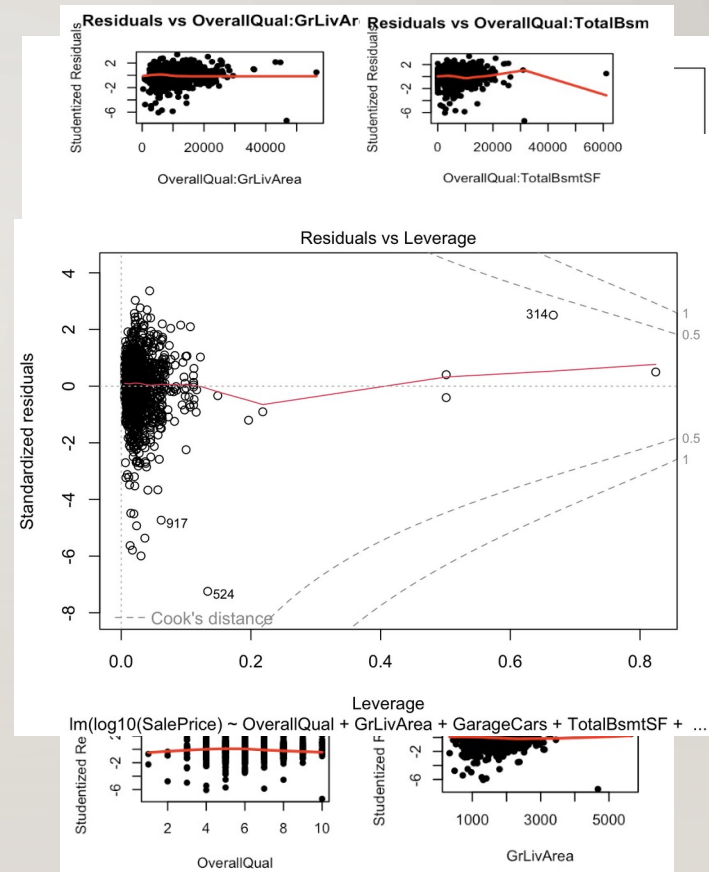

MODEL DIAGNOSTIC

Residuals vs. Fitted plot: improved homoscedasticity, with observations 31, 63, and 524 as outliers
Q-Q plot: lighter right tail and heavy left tail remains.

Residuals against each predictor:

- Non-linear, curved patterns in GrLivArea and TotalBsmtSF
- Interaction shows non-linear pattern

Residuals vs. Leverage: no points exhibit a high Cook's Distance



RELIEVING THE ASSUMPTION VIOLATION

Log10(x + 1), to both LotArea and TotalBsmtSF, along with all corresponding interaction terms

```
model_final_3 <- lm(log10(SalePrice) ~ OverallQual + GrLivArea + GarageCars + log10(TotalBsmtSF+1)
+ YearBuilt + Neighborhood + FullBath + KitchenQual + log10(1+LotArea)
+ OverallQual:GrLivArea + OverallQual:log10(TotalBsmtSF+1)
+ GrLivArea:log10(TotalBsmtSF+1) + I(GarageCars^2))
```

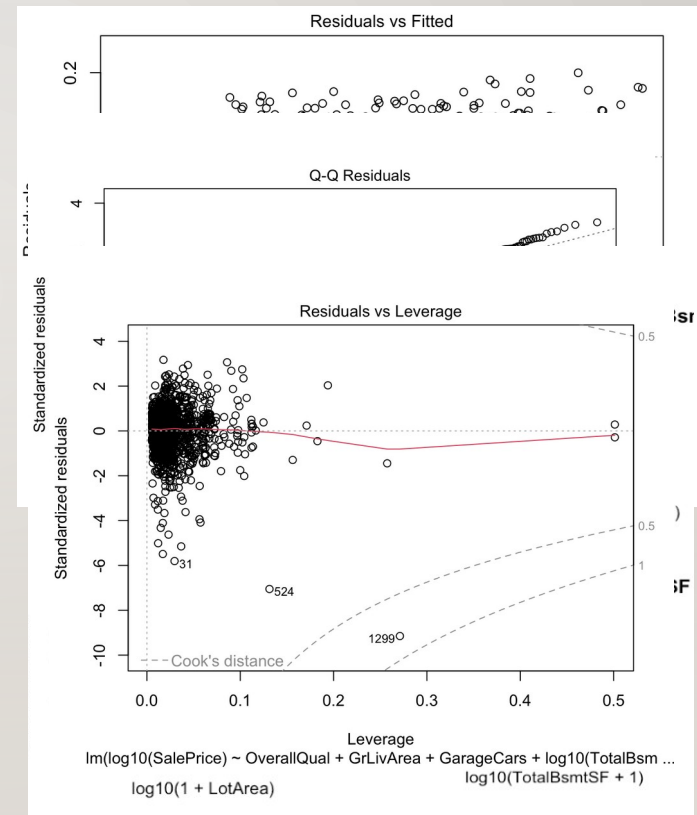
MODEL DIAGNOSTIC

Residuals vs. Fitted plot: maintains consistent random scatter

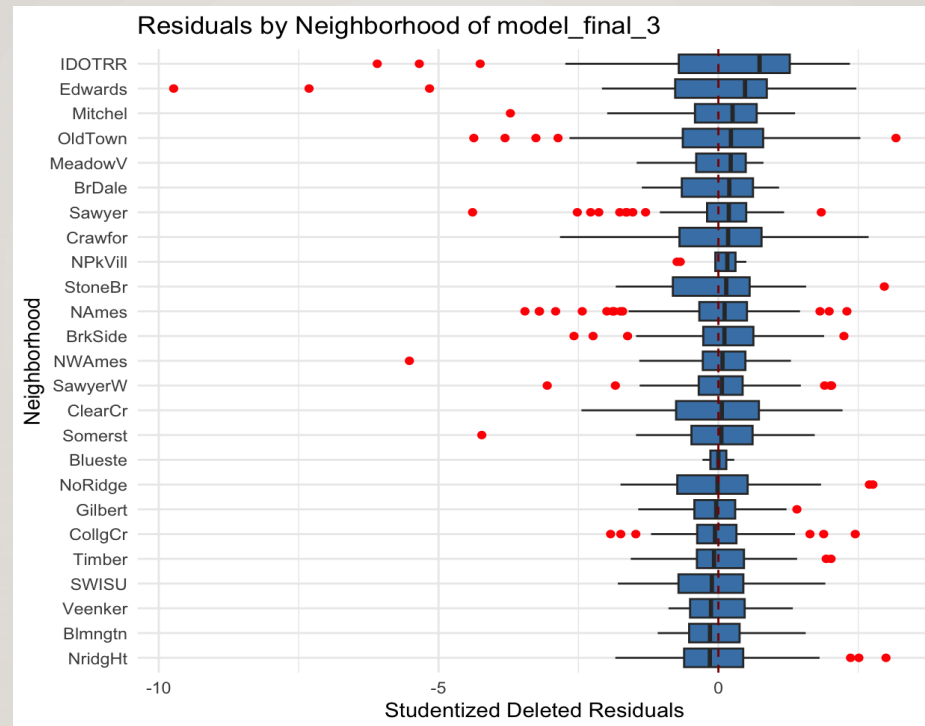
Q-Q plot: even more lighter right tail, but left tail is heavier

Residuals against each predictor: show a random distribution

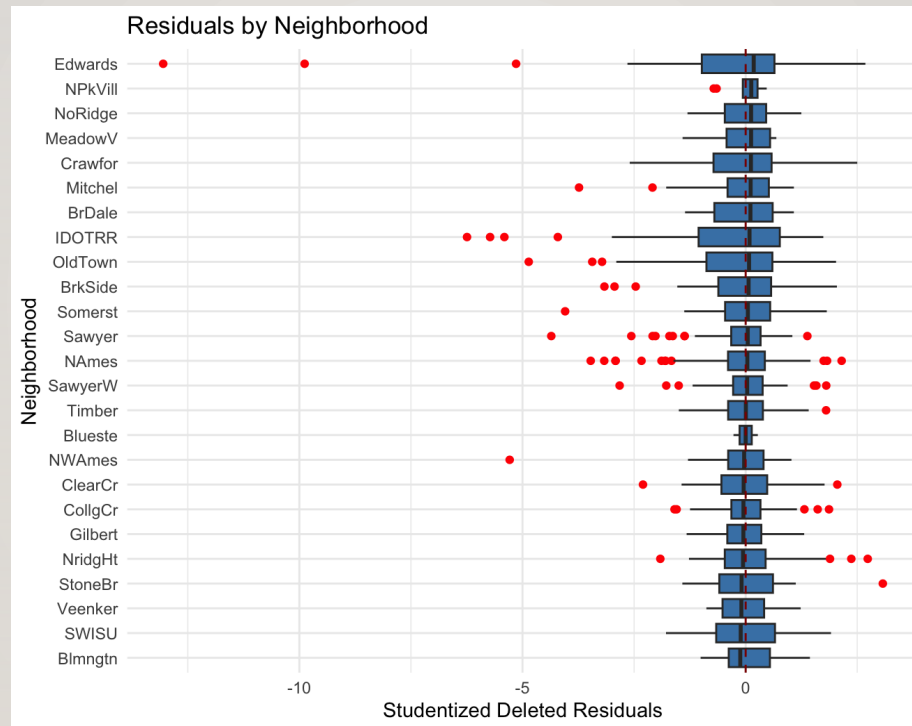
Residuals vs. Leverage: point 1299 remains the sole observation with a high Cook's Distance



NON-CONSTANT RESIDUALS BETWEEN GROUP



ROBUST LINEAR MODEL



INFLUENCE ANALYSIS

House 1299, studentized deleted residual of -9.74 . This is an unusual luxury house with Overall Quality of 10 and huge areas, but sold at a very cheap price.

House 524, studentized deleted residual of -7.31 .

House 31, studentized deleted residual of -6.09 .

House 496, studentized deleted residual of -5.34 .

```
> dat_clean[all_criteria_rows_3 & abs(res3) > 3 , ]$res3
```

31	186	384	411	463	496	524	633	659	667
-6.095453	<u>3.173371</u>	-3.058570	-5.161280	-4.392402	-5.344099	-7.314251	-5.521187	-3.197047	-3.456660
729	813	875	899	917	969	1063	1299	1325	1454
-3.199817	-4.255876	-3.260505	<u>2.995083</u>	-2.733919	-4.369630	-3.812274	-9.733666	-4.227709	-3.716573

OUTLIERS: ROBUST VS OLS

	SalePrice	predicted_lm	predicted_robust	Difference_lm	Difference_robust
1	40000	92324.42	102026.82	-52324.42	-62026.82
2	475000	309522.66	386958.08	165477.34	88041.92
3	76000	118784.74	116389.76	-42784.74	-40389.76
4	60000	124299.26	130366.91	-64299.26	-70366.91
5	62383	116881.26	120513.14	-54498.26	-58130.14
6	34900	73101.05	82459.44	-38201.05	-47559.44
7	184750	483142.66	781083.92	-298392.66	-596333.92
8	82500	182924.27	183065.15	-100424.27	-100565.15
9	97500	157345.55	157681.23	-59845.55	-60181.23
10	129000	214830.04	218326.56	-85830.04	-89326.56
11	110000	172633.96	171118.92	-62633.96	-61118.92
12	55993	97977.56	105430.72	-41984.56	-49437.72
13	66500	104834.10	108305.48	-38334.10	-41805.48
14	611657	386232.11	404122.83	225424.89	207534.17
15	35311	63087.24	79409.55	-27776.24	-44098.55
16	37900	73966.59	78879.53	-36066.59	-40979.53
17	90000	153068.41	151427.08	-63068.41	-61427.08
18	160000	501650.20	1014201.00	-341650.20	-854201.00
19	147000	266621.63	271039.55	-119621.63	-124039.55
20	84500	141913.46	148383.84	-57413.46	-63883.84

Coefficient Interpretation

Model Evaluation Metrics

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.205e+00	2.759e-01	7.990	2.76e-15	***
OverallQual	1.204e-02	1.114e-02	1.081	0.279820	
GrLivArea	2.792e-04	2.735e-05	10.210	< 2e-16	***
GarageCars	2.338e-02	7.966e-03	2.935	0.003393	**
log10(TotalBsmtSF + 1)	1.909e-02	1.495e-02	1.277	0.201924	
YearBuilt	1.041e-03	1.309e-04	7.949	3.82e-15	***
NeighborhoodBlueste	1.781e-02	4.800e-02	0.371	0.710616	
NeighborhoodBrDale	-4.228e-02	2.322e-02	-1.821	0.068829	.
NeighborhoodBrkSide	4.914e-03	2.008e-02	0.245	0.806656	
NeighborhoodClearCr	2.260e-02	2.178e-02	1.038	0.299543	
NeighborhoodCollgCr	-1.036e-02	1.712e-02	-0.605	0.545189	
NeighborhoodCrawfor	6.578e-02	1.995e-02	3.297	0.001003	**
NeighborhoodEdwards	-3.539e-02	1.866e-02	-1.896	0.058137	.
NeighborhoodGilbert	-2.301e-02	1.817e-02	-1.266	0.205561	
NeighborhoodIDOTRR	-7.782e-02	2.134e-02	-3.647	0.000275	***
NeighborhoodMeadowV	-2.872e-02	2.307e-02	-1.245	0.213489	
NeighborhoodMitchel	-1.837e-02	1.922e-02	-0.956	0.339358	
NeighborhoodNames	-3.234e-03	1.789e-02	-0.181	0.856607	
NeighborhoodNoRidge	5.232e-02	1.965e-02	2.663	0.007841	**
NeighborhoodNPkVill	6.916e-03	2.677e-02	0.258	0.796209	
NeighborhoodNrIdght	3.679e-02	1.792e-02	2.053	0.040228	*
NeighborhoodNWAmes	-1.294e-02	1.850e-02	-0.700	0.484297	
NeighborhoodOldTown	-3.774e-02	1.953e-02	-1.932	0.053502	.
NeighborhoodSawyer	-9.578e-03	1.898e-02	-0.505	0.613887	
NeighborhoodSawyerW	-2.239e-02	1.848e-02	-1.212	0.225886	
NeighborhoodSomerst	6.938e-03	1.724e-02	0.403	0.687364	
NeighborhoodStoneBr	5.708e-02	2.052e-02	2.782	0.005474	**
NeighborhoodSWISU	-7.559e-03	2.255e-02	-0.335	0.737556	
NeighborhoodTimber	4.201e-04	1.976e-02	0.021	0.983038	
NeighborhoodVeenker	4.745e-02	2.564e-02	1.851	0.064420	.
FullBath	-2.911e-04	4.796e-03	-0.061	0.951614	
KitchenQualFa	-1.032e-01	1.425e-02	-7.245	7.06e-13	***
KitchenQualGd	-4.372e-02	8.300e-03	-5.267	1.60e-07	***
KitchenQualTA	-7.182e-02	9.269e-03	-7.748	1.77e-14	***
log10(1 + LotArea)	1.192e-01	1.130e-02	10.554	< 2e-16	***
I(GarageCars^2)	4.324e-05	2.297e-03	0.019	0.984983	
OverallQual:GrLivArea	-1.157e-05	2.292e-06	-5.049	5.01e-07	***
OverallQual:log10(TotalBsmtSF + 1)	1.342e-02	3.620e-03	3.706	0.000219	***
GrLivArea:log10(TotalBsmtSF + 1)	-3.748e-05	9.299e-06	-4.031	5.85e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Coefficient Interpretation

Largest Positive and Negative Impacts by Neighborhood

Neighborhood	Coefficient
IDOTRR	-0.0778
BrDale	-0.0423
OldTown	-0.0377
Edwards	-0.0354
MeadowV	-0.0287
NridgHt	0.0368
Veenker	0.0475
NoRidge	0.0523
StoneBr	0.0571
Crawfor	0.0658

In Conclusion...

- **Overall Quality (OverallQual)** is the strongest predictor of housing prices, with the highest correlation with $\log(\text{SalePrice})$
- **Living area (GrLivArea), basement size (TotalBsmtSF), garage capacity (GarageCars), and lot area (LotArea)** are also strongly associated with price, suggesting that each of these variables individually contribute to price. A property with lots of land and lots of living area will tend to cost more.
- Location, represented by **Neighborhood**, has a significant impact on housing prices even after controlling for structural characteristics. Some neighborhoods are associated with higher housing prices (including StoneBr, NoRidge, Crawfor,...) while others (IDOTRR, OldTown) are associated with lower prices.
- **The final model achieves a high adjusted $R^2 \approx 0.875$** , indicating that it explains a large proportion of the variation in housing prices.
- After applying appropriate transformations (e.g., log transformation), the model satisfies key assumptions more closely, including improved linearity, normality, and constant variance. Therefore, **multiple linear regression is an effective and appropriate method for modeling housing prices in this dataset.**

Future Considerations

- **Making predictions with real data and analyze accuracy**
- **Add a variable to adjust for present year**
- **continuous improvement...**